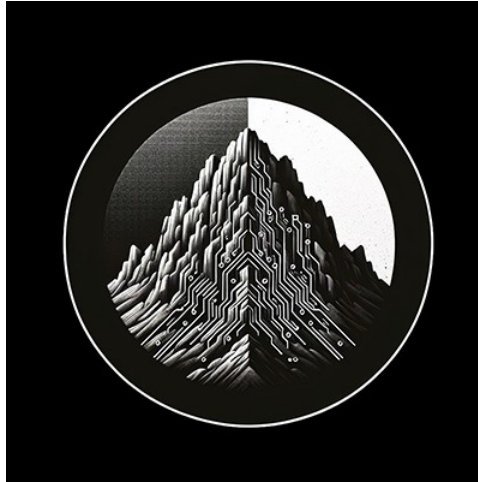




RTL Design Sherpa

DDR2/LPDDR2 Characterization Nexys A7 Project Guide 0.90

July 2026

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1 Document Information

This is the operator / developer guide for the Nexys A7 **DDR2/LPDDR2 Characterization** project (projects/NexysA7/ddr2-characterization). It explains what the project characterizes, how the harness works, how to build / simulate / program / run it, and how the harness CSR is configured.

The memory controller under test — **pumice** — is separate IP (projects/components/memory-controllers/pumice-ddr2-lpddr2). This guide treats it as a **black box** driven by the characterization harness; it does not document the controller’s internals.

Note: the project’s top-level README.md still carries an early “skeleton — RTL not yet written” status line. That is stale: the harness, host, sim, and both flows are built and running. Trust this guide and the RTL/host sources over that status.

1.1 References

Reference documents

Source	Title
RTL Design Sherpa	build-perf/host/ADDRESS_MAP.md (bridge + CSR map)
RTL Design Sherpa	build-perf/bin/README_a7ddrphy.md (PHY regen)
RTL Design Sherpa	flows-litedram-uart/HARNESS_PLAN.md (baseline flow)
RTL Design Sherpa	bin/TBClasses/harness/ (shared UART-char collateral)
Micron	MT47H64M16HR-25E DDR2 SDRAM datasheet
ARM	AMBA AXI4 / AXI4-Lite / APB Protocol Specifications

1.2 Terminology

DFI

DDR PHY Interface. The command/data protocol between the memory controller and the PHY. Here pumice emits a phase-packed flat DFI v2.1 bus.

a7ddrphy

LiteDRAM's Artix-7 DDR2 PHY (4:1 serdes, 800 Mbps). Uses Xilinx OSERDESE2/ISERDESE2/IDELAYE2 and therefore does **not** simulate in Verilator.

GEAR / DFI_RATE

The DFI phase count. $\text{GEAR_RATIO} = \log_2(\text{DFI_RATE})$. Must match between the RTL build and the sim model or reads corrupt.

Engine

A master-side AXI4 traffic generator: the write engine emits an LFSR data pattern; the read engine re-reads and CRC-checks it.

Leveling

Firmware-driven a7ddrphy read/write training over the PHY-CSR passthrough window; there is no hardware leveling FSM.

1.3 Revision History

Revision history

Version	Date	Notes
0.90	2026-07-18	Initial project guide

2 Overview

2.1 What it characterizes

This project performs end-to-end, board-level **performance, latency, and data-integrity characterization** of a DDR2/LPDDR2 memory controller on real silicon. It drives the onboard Micron MT47H64M16HR-25E DDR2 (x16, single-rank, 128 MiB, up to 800 Mbps) through the controller against the real device, and reports throughput (MB/s), latency histograms, and per-workload CRC integrity across burst-length / stride / gap / page-policy / scheme sweeps.

There is no soft CPU in the FPGA — an off-board Python **host** drives the harness over the FTDI USB-UART (115200 8N1). The same host programs run against a cocotb simulation (Chapter 6).

2.2 The three flows

The project is organized as sibling “flows” plus a shared framework:

The characterization flows

Flow	Controller (DUT)	What it is
build-perf/	pumice DDR2 (black box) + DFI + a7ddrphy	Primary flow: UART-driven host + Vivado build of the pumice harness on the Nexys A7.
flows-litedram-uart/	LiteDRAM litedram_core	Apples-to-apples baseline: LiteDRAM’s own DDR2 controller (self-initializing, own PLL/PHY) driven by the same engines/taps/host.
ddr2_char_framework/	—	Shared instrumentation: harness RTL, the engine wrapper (ddr2_char_macro), the 1→4 AXIL bridge, and the cocotb DV.

The naming convention is flows-<controller>-uart/ for on-Nexys builds (host over UART, no CPU). pumice is separate IP under projects/components/memory-controllers/pumice-ddr2-lpddr2/.

2.3 Board and tools

Board and tool requirements

Item	Detail
Board	Digilent Nexys A7-100T, Artix-7 xc7a100tcsg324-1

Item	Detail
DRAM	Onboard Micron MT47H64M16 DDR2 x16, ROW_WIDTH = 13
Build	Vivado (batch mode)
Host	Python 3 + pyserial
Sim	Verilator + cocotb / cocotb_test / pytest
PHY regen (only)	Python 3.10 + migen / litex 2024.12 / litedram 2024.12

2.4 Workload families

The characterization sweeps four access-pattern families (build-perf/host/pumice_char.py):

Access-pattern families

Family	Access	Purpose
incremental	contiguous linear	best-case sequential
row_major	bounded to one page	guaranteed page HIT
col_major	stride = one row, same bank	guaranteed page MISS (worst case)
col_major_interleave d	stride = one bank	activates the bank pipeline

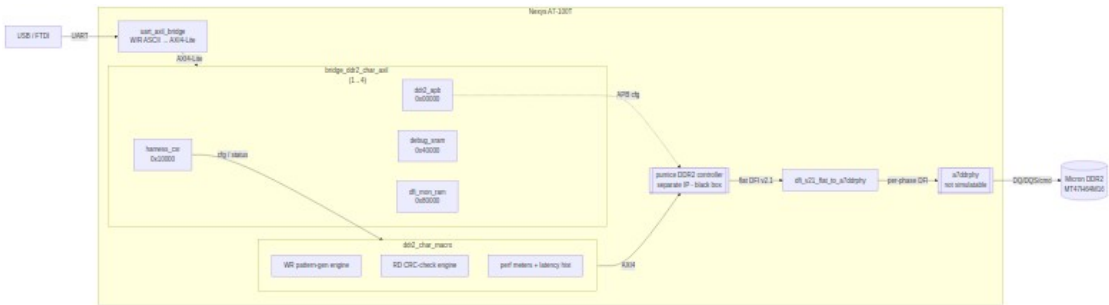
Controller-config presets (baseline, bank_interleave, open_page, inorder, reorder, happy_hybrid, fast_refresh, slow_refresh) each flip one lever from baseline; they are driven over the pumice APB slave (black box). Run profiles smoke / matrix / full are shared by sim and board.

3 How It Works

3.1 The transport spine

Host commands reach the DUT over one UART link that fans out to four AXI4-Lite slaves:

3.1.1 Figure 2.1: DDR2 Characterization Harness Spine



DDR2 characterization harness spine

Source: [01_harness_spine.mmd](#)

The UART wire protocol is ASCII, line-based, 115200 8N1: a write is W <hex-addr32> <hex-data32>\n → the FPGA replies OK\n; a read is R <hex-addr32>\n → replies 0x<hex-data32>\n. The bridge RTL is projects/components/converters/rtl/uart_to_axil4/uart_axil_bridge.sv; the host framing is UARTAxiBridge (projects/components/converters/bin/uart_axi_bridge.py).

3.2 The 1→4 AXIL bridge

bridge_dds2_char_axil is generated from ddr2_char_framework/rtl/bridges/configs/bridge_dds2_char_axil.toml (one master host, four slaves). It decodes the four windows in Chapter 5 and routes each host transaction to the right slave via per-slave adapters.

3.3 Key RTL blocks

Key RTL blocks

File	Role
build-perf/rtl/ddr2_char_top.sv	FPGA pin-level top: wraps the harness, the flat-DFI→per-phase adapter, and the a7ddrphy black box; MMCM clock synthesis (sys / sys2x / sys4x / sys4x_dqs), IDELAYCTRL, DDR2 pads.
build-perf/rtl/ddr2_char_harness.sv	Internal integration: uart_axil_bridge + bridge_dds2_char_axil + harness_csr + debug_sram + dfi_mon_ram + ddr2_char_macro +

File	Role
	char timer + LED / 7-seg.
ddr2_char_framework/rtl/ ddr2_char_macro.sv	Binds the two AXI4 engines to pumice's s_axi, and holds the perf taps (bus meters + latency histogram) on the internal AXI wires.
ddr2_char_framework/rtl/ harness_csr.sv	The AXIL CSR slave (Chapter 5). Hand-written (self-clearing pulses, latches, PHY passthrough).
ddr2_char_framework/rtl/ dfi_v21_flat_to_a7ddrphy.sv	Combinational adapter: pumice's phase-packed flat DFI v2.1 → a7ddrphy per-phase ports (NPHASES = 4).
ddr2_char_framework/rtl/ a7ddrphy_stub.sv	Port-shape black box of a7ddrphy for Verilator (real body swapped in by Vivado).

3.4 The characterization engines

Inside ddr2_char_macro, two master-side AXI4 engines generate and check traffic against the controller's s_axi port:

- **Write engine** (axi4_master_wr_pattern_gen) — emits an LFSR data pattern across the programmed address pattern, and latches an expected CRC.
- **Read engine** (axi4_master_rd_crc_check) — re-reads the same pattern and computes an actual CRC, plus a per-beat mismatch count.

Alongside them, **perf taps** — axi_bus_meter (write and read) and axi_perf_latency_hist — observe the internal AXI wires to produce the producer/backpressure/starve/idle buckets and the latency histogram read back in Chapter 5.

3.5 The controller, DFI, and PHY (black boxes)

The controller (**pumice**) sits inside ddr2_char_macro between the AXI engines and the flat DFI output. Its configuration registers are reached over the ddr2_apb bridge slave at base 0x0. This guide documents only that the harness drives it — not its internals.

pumice's flat DFI v2.1 bus is adapted to the a7ddrphy per-phase interface, then to the DDR2 pads. **a7ddrphy does not simulate in Verilator** (it uses Xilinx SERDES/IDELAY primitives); the sim connects at the DFI level with a behavioral model instead (Chapter 6). On hardware, a7ddrphy

calibration is driven by firmware (the host) over the PHY-CSR passthrough window — there is no hardware leveling FSM.

4 Test Methodology

This chapter describes what is exercised on the FPGA block (Figure 2.1) and how throughput, latency, and data integrity are measured.

4.1 What is under test

The **pumice** memory controller (black box) driving the real onboard DDR2, stimulated by the two master-side AXI4 engines. Every measurement is taken on the internal AXI wires between the engines and the controller, so the numbers reflect the controller + PHY + device as a whole.

4.2 What is measured, and how

What is measured

Quantity	How it is obtained
Throughput	$\text{beats} \times \text{bytes_per_beat} / (\text{TIMER_CYCLES} \times 10 \text{ ns})$ — the timer bounds the run in <code>sys_clk</code> cycles
Latency histogram	OBS_HIST_SEL selects bus (rd/wr) + metric (AR→firstR / AR→RLAST) + bin; OBS_HIST_COUNT / OBS_HIST_TOTAL read it back. Bin <i>b</i> covers $[2^b, 2^{(b+1)})$ cycles
Data integrity	CRC_ACTUAL vs CRC_EXPECTED (CRC_MATCH), plus BEATS_MISM
Utilization	OBS_{RD,WR}_{PROD,BP,STARV,IDLE} bus-meter buckets

A measurement run: program the engine config (address pattern, burst length, txn count, LFSR seed), `clear_stats`, kick with `CTRL.start_wr / start_rd`, poll `STATUS.wr_done / rd_done`, then read the timer, CRC, histogram, and meters. The write engine latches the expected CRC as it generates the pattern; the read engine re-reads and computes the actual CRC.

4.3 Workloads

Characterization workloads

Axis	Values
Access pattern	<code>incremental</code> , <code>row_major</code> , <code>col_major</code> , <code>col_major_interleaved</code>

Axis	Values
Controller preset	baseline, bank_interleave, open_page, inorder, reorder, ... (over the pumice APB slave)
Burst / count / stride	swept per --char-profile (smoke / matrix / full)

4.4 The oracle

Integrity is judged against a deterministic **LFSR pattern + CRC-32**: the write engine's CRC_EXPECTED must equal the read engine's CRC_ACTUAL with BEATS_MISM = 0. The same golden logic runs in sim (Chapter 6).

4.5 Measurement pitfalls

- STATUS.any_error is **sticky** — clear it with clear_stats between phases or every later run looks wedged.
- Keep board runs $\text{txn} \leq 1024$ (read-engine CAM ceiling) for clean completion; the *rate* stays valid at scale.
- col_major stride $\times$ large txn can alias past the 128 MB device — bound it.
- The sim must be built at the board's exact DFI_RATE / GEAR or it passes while the board fails (Chapter 6).

5 Build and Run

All commands assume the Python environment is sourced (sets REPO_ROOT, SIM=verilator, PATH, PYTHONPATH):

```
cd /path/to/RTLDesignSherpa && source env_python
cd projects/fpga-systems/NexysA7/pumice/build-perf
```

5.1 build-perf make targets

UART ?= /dev/ttyUSB1, BAUD ?= 115200, VIVADO ?= vivado.

build-perf make targets

Target	Action
make lint	Verilator lint of ddr2_char_top via the harness filelist
make project / make synth	create Vivado project / synth-only +

Target	Action
	reports
make bitstream	full synth + impl + bitgen (~10–30 min) → bitstream/ddr2_char.bit
make bitstream-ila	same, plus an ILA on the DFI boundary → .bit + .ltx
make program	flash the board
make utilization/make timing	print the latest reports
make smoke	host/run_smoke.py — link + one linear WR/RD integrity pass
make status	host/ddr2_char.py — one-shot BUILD_ID + status dump
make level	host/pumice_master.py --level- only — a7ddrphy leveling
make simple	host/pumice_master.py --simple — init + one write→read pass
make characterize	host/pumice_master.py --full — full workload sweep
make sweep-rddly/make train- deskew	PHY timing sweeps
make host-test	pytest host/test_pumice_master.py (mock UART, no board)
make sim/sim-smoke/sim-simple/ sim-level	run the same host programs in cocotb sim
make clean/clean-all	remove build artifacts

The other Makefiles: flows-litedram-uart/Makefile (make regen/make bitstream/make program for the LiteDRAM baseline) and ddr2_char_framework/dv/tests/Makefile (the cocotb macro suite: make run, run-smoke, run-<shape>, and parallel run-{gate,func,full}-parallel).

5.2 The host programs (build-perf/host/)

The port defaults to `--port auto`, which probes every `/dev/ttyUSB*` and keeps the board that answers `BUILD_ID == 0x44445232` (“DDR2”).

run_smoke.py — the first thing to run after flashing. Programs a linear WR+RD workload, kicks both engines, and checks CRC / cycles / perf.


```
python3 host/run_smoke.py --port /dev/ttyUSB1 --txn 1024 --blen 8 --seed 0xDEADBEEF
```

Pass = BUILD_ID matches, both engines done, CRC_ACTUAL == CRC_EXPECTED, BEATS_MISMATCHED == 0, and TIMER.pass.

pumice_master.py — the orchestration program; one mode is required:

```
python3 host/pumice_master.py --port /dev/ttyUSB1 --level-only # PHY read/write leveling
python3 host/pumice_master.py --port /dev/ttyUSB1 --simple # init + one write→read pass
python3 host/pumice_master.py --port /dev/ttyUSB1 --full # full workload sweep
python3 host/pumice_master.py --port /dev/ttyUSB1 --char --char-profile matrix --char-scale 1000 --csv out.csv
```

Useful options: --no-level, --rd-phase, --rd-delay, --char-configs, --char-scale N (≈1000 on FPGA), --level-cache JSON, --clk-mhz.

ddr2_char.py — the driver library plus a one-shot status CLI (python3 host/ddr2_char.py --port auto). It exports DDR2CharDriver, the base-address constants, and the enum constants used throughout.

5.3 Typical bring-up sequence

```
make bitstream
make program
make smoke UART=/dev/ttyUSB1 # link + integrity
make level # find the read/write eye
make simple # one clean write→read
make characterize # full sweep -> CSV
# no board? prove the same programs in sim:
make sim
```

6 Harness CSR Configuration

6.1 Bridge address map

The UART bridge fans out to four AXI4-Lite slaves. Bases come from bridge_ddr2_char_axil.toml (mirrored in build-perf/host/ADDRESS_MAP.md and as constants in ddr2_char.py):

Bridge slave address map

Constant	Slave	Base	Size	Protocol
DDR2_APB_BASE	ddr2_apb	0x0000_00	4 KB	APB — pumice

Constant	Slave	Base	Size	Protocol
		00		controller CSR (black box)
HARNESS_CSR_BASE	harness_csr	0x0001_00	4 KB	AXIL — this chapter
DEBUG_SRAM_BASE	debug_sram	0x0004_00	256 KB	AXIL 64b — MonBus/DFI trace ring
DFI_MON_RAM_BASE	dfi_mon_ram	0x0008_00	4 KB	AXIL — DFI cmd observability

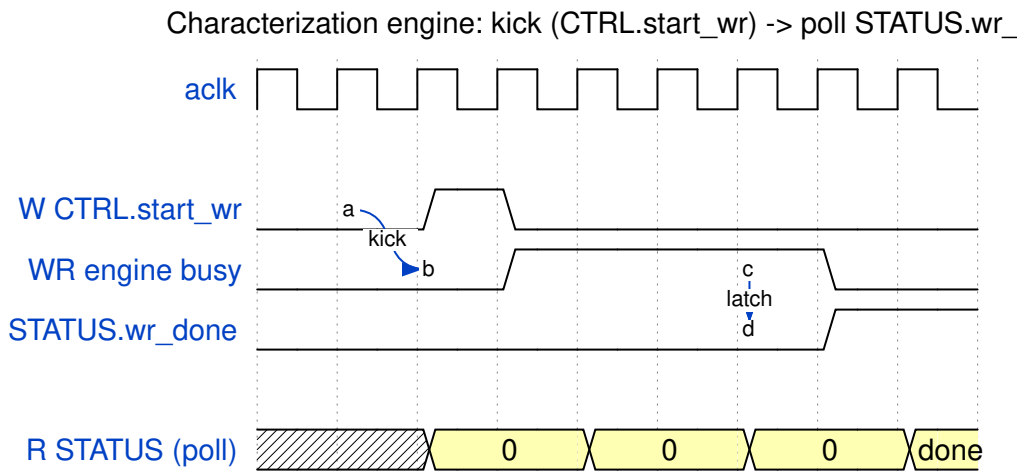
6.2 By-name access (RDL → regmap)

`harness_csr.sv` is hand-written; `harness_csr.rdl` is a descriptor of it, from which the by-name host regmap is generated. Registers are accessed by name via `UartRegisterMap`, never by hardcoded offset. Regenerate after editing either:

```
python3 bin/peakrdl_generate.py \
projects/NexysA7/ddr2-characterization/ddr2_char_framework/rtl/harness
_csr.rdl \
--regmap --docs-only --no-html --no-markdown \
--regmap-output
projects/NexysA7/ddr2-characterization/ddr2_char_framework/dv/tbclasses/
harness_csr_regmap.py
```

A consistency test (`build-perf/host/test_harness_regmap_consistency.py`) asserts the generated regmap matches the SV header table.

6.2.1 Waveform 5.1: Engine Kick and Poll



Engine kick and poll

Source: [01_engine_kick.json](#)

6.3 Control / status (offsets relative to HARNESS_CSR_BASE)

Control / status registers

Register	Offset	Field	Bits	Access	Description
CTRL	0x00	start_w r	[0]	W	Pulse: kick the write engine
CTRL	0x00	start_r d	[1]	W	Pulse: kick the read engine
CTRL	0x00	clear_s tats	[2]	W	Pulse: clear perf + sticky errors
CTRL	0x00	freeze_ trace	[3]	RW	Latch: freeze perf counters
CTRL	0x00	soft_re set	[4]	W	Pulse: soft reset (does not clear sticky errors)
STATUS	0x04	wr_done / rd_done	[0] / [1]	R	Engine completion
STATUS	0x04	wr_erro r /	[2] / [3]	R	Per-engine error

Register	Offset	Field	Bits	Access	Description
STATUS	0x04	rd_error any_error	[4]	R	Sticky OR of errors — cleared by clear_stats
STATUS	0x04	init_done / init_fail	[6] / [7]	R	Controller init status
DBG_WR_PTR	0x08	value	[31:0]	R	Words written to debug_sram since clear
DBG_OVERFLOW	0x0C	overflow	[0]	R	Sticky trace-overflow
CRC_EXPECTED	0x10	value	[31:0]	R	Write-engine expected CRC
CRC_ACTUAL	0x14	value	[31:0]	R	Read-engine actual CRC
CRC_MATCH	0x18	match	[0]	R	CRC_ACTUAL == CRC_EXPECTED
CRC_MATCH	0x18	exp_val id / act_val id	[1] / [2]	R	CRC captured
CRC_MATCH	0x18	beats_mism_nz	[3]	R	Any beat mismatched
SCRATCH	0x1C	value	[31:0]	RW	Host link ping
BUILD_ID	0x20	value	[31:0]	R	0x44445232 (“DDR2”)
BEATS_MISM	0x24	value	[31:0]	R	Beats mismatched

6.4 Timer

Characterization timer registers

Register	Offset	Bits	Access	Description
TIMER_CTRL	0x28	[0] clear	W	Pulse: clear the cycle

Register	Offset	Bits	Access	Description
TIMER_STATUS	0x2C	[0] done, [1] running, [2] pass	R	counter Timer state
TIMER_CYCLES_LO/HI	0x30 / 0x34	[31:0]	R	64-bit cycle count (10 ns/cycle)
TIMER_EXP_BEATS	0x38	[31:0]	RW	Beat-count stop trigger (0 = disable)
TIMER_{R,W}_{FIRST, LAST}_{LO, HI}	0x40–0x5C	[31:0]	R	First/last R/W beat timestamps (64-bit each)

6.5 Runtime controller cfg + DFI tuning

Runtime controller config and live DFI tuning

Register	Offset	Field	Bits	Access	Description
CTRLR_CFG	0x60	memtype	[0]	RW	0 = DDR2, 1 = LPDDR2
CTRLR_CFG	0x60	t_phy_wrlat	[15:8]	RW	PHY write latency (see Chapter 7 — board wants 0)
CTRLR_CFG	0x60	t_rddata_en	[23:16]	RW	Read-data-enable latency
CTRLR_CFG	0x60	rd_in_order	[24]	RW	Force in-order reads
CTRLR_CAP	0x64	cap_lookahead_max	[3:0]	RW	Advertised OOO look-ahead depth
CTRLR_CAP	0x64	cap_synth_mask	[7:4]	RW	Advertised feature mask
DFI_TUNING	0x68	cmd_delay	[3:0]	RW	Live DFI cmd delay (reset 1) — no rebuild
DFI_TUNING	0x68	rddata_	[7:4]	RW	Live DFI read-data

Register	Offset	Field	Bits	Access	Description
		delay			delay (reset 0)

6.6 a7ddrphy calibration passthrough (hardware only)

Indirect access to the a7ddrphy's read/write-leveling knobs; firmware-driven, no HW FSM.
Meaningless in sim (the stub ignores it). Knob map:
rtl-vivado/a7ddrphy/a7ddrphy_csr_map.txt.

a7ddrphy calibration passthrough

Register	Offset	Bits	Access	Description
PHY_CSR_ADDR	0x80	[9:0]	RW	a7ddrphy CSR word index
PHY_CSR_WDATA	0x84	[31:0]	RW	Value to write
PHY_CSR_CTRL	0x88	[0] pulse	W	Drive one CSR-bus write
PHY_CSR_RDATA	0x8C	[31:0]	R	dat_r for the current PHY_CSR_ADDR

6.7 Engine config (WR 0x100–0x128, RD 0x180–0x1A8)

The read block mirrors the write block at +0x80.

Write-engine config (read engine identical at +0x80)

Register	Offset	Field	Bits	Description
WR_START_ADDR	0x100	value	[31:0]	Start address
WR_STRIDE_0/1	0x104 / 0x108	value	[23:0]	Signed strides
WR_WRAP_MASK_0 /1	0x10C / 0x110	value	[31:0]	Address wrap masks
WR_BLEN_TXN	0x114	burst_len	[7:0]	AXI burst length
WR_BLEN_TXN	0x114	txn_count	[23:8]	Transaction count
WR_BLEN_TXN	0x114	gap	[27:24]	Inter-burst gap
WR_AXI_ATTR	0x118	axi_id	[7:0]	AXI ID
WR_AXI_ATTR	0x118	id_mode	[9:8]	0 fixed / 1 counter / 2 LFSR

Register	Offset	Field	Bits	Description
WR_AXI_ATTR	0x118	axi_size	[12:10]	AXI size (AXI_SIZE_8 = 3)
WR_AXI_ATTR	0x118	axi_burst	[14:13]	0 fixed / 1 incr / 2 wrap
WR_AXI_ATTR	0x118	data_mode	[15]	Data pattern mode
WR_LFSR_SEED	0x11C	value	[31:0]	Pattern seed (must match RD for CRC)
WR_HASH_SEED0/ 1/2	0x120–0x128	value	[31:0]	Hash seeds

6.8 Perf observability (0x1C0–0x1E8)

All 32-bit; cleared by CTRL.clear_stats, frozen by CTRL.freeze_trace.

Perf observability (bin b covers $[2^b, 2^{(b+1)})$ cycles)

Register	Offset	Field	Bits	Description
OBS_RD_{PROD,BP,STARV,IDLE}	0x1C0–0x1CC	value	[31:0]	Read data-channel meter buckets
OBS_WR_{PROD,BP,STARV,IDLE}	0x1D0–0x1DC	value	[31:0]	Write meter buckets
OBS_HIST_SEL	0x1E0	bus	[0]	0 = read, 1 = write
OBS_HIST_SEL	0x1E0	metric	[1]	0 = AR→firstR / AW→B, 1 = AR→RLAST
OBS_HIST_SEL	0x1E0	bin	[5:2]	Histogram bin index (0–15)
OBS_HIST_COUNT	0x1E4	value	[31:0]	Count in the selected bin
OBS_HIST_TOTAL	0x1E8	value	[31:0]	Total txns on the selected metric

7 Simulation / Silicon Equivalence

This project follows the shared UART-characterization methodology (bin/TBClasses/harness/): **one authored-once host program runs byte-for-byte identically on the FPGA and in a cocotb simulation.** The equivalence boundary is the ASCII W/R byte stream on the UART wire; only the

bottom ByteChannel differs — SerialChannel (pyserial) on the board vs CocotbUartChannel on the DUT pins in sim.

7.1 Testbenches

Simulation testbenches

tb_top	Wraps	Drives cfg via
ddr2_char_uart_tb_top.sv	the full ddr2_char_harness (real UART bridge + harness_csr + 1→N bridge + engines + pumice)	the real UART pins (cocotb UARTMaster/Monitor)
ddr2_char_macro_tb_top.sv	the inner ddr2_char_macro	direct ports + an APB BFM

The UART tb is the equivalence vehicle: it exposes real UART pins so the sim consumes the identical byte stream the host sends to silicon. The DFI side is wired to internal phy_dfi_* nets so the framework's DFISlavePHY + MemoryModel loopback auto-binds by prefix. **There is no a7ddrphy in sim** — it reproduces the digital DFI handshake, not the analog eye.

7.2 How the same program runs in sim

test_ddr2_char_uart.py builds the DFISlavePHY + MemoryModel backend, then:

```
r = await cocotb.external(prog)() # prog is an UNMODIFIED
pumice_master routine
```

make_uart_channel(dut, clock, clks_per_bit) returns a CocotbUartChannel that drives the UART pins; the synchronous host program runs in a worker thread under cocotb.external, and the channel's read/write are cocotb.function wrappers that step the sim. The baud is lowered for sim (UART_BAUD = 6_250_000 → CLKS_PER_BIT = 16 vs 868 on board) — bit-timing only, the byte stream is unchanged. make_sim / sim-smoke / sim-simple / sim-level run the same programs the board runs.

7.3 Match the board's exact DFI config

GEAR_RATIO = log2(DFI_RATE) and **must** match the compile-time DFI_RATE or reads corrupt. The test threads TEST_DFI_RATE / TEST_GEAR_RATIO / TEST_DRAM_BEAT_BYTES into both the SV parameters={...} and the host via drv.set_dfi_phase(...). The board is built at DFI_RATE

= 4 (GEAR = 2, AXI 64 / DRAM beat 32); a rate-2 / GEAR-1 sim will **pass while the board fails** — it skips the very paths that break. Named variants (..._smoke_rate4, ..._smoke_rate4_x16, ..._pagehit_rate4_x16, ...) exist to run the board's real geometry.

7.4 What the sim can and cannot reproduce

- **CAN:** everything digital — the controller/engines, the bridge RTL, DFI-level handshakes, register logic. This is where it earns its keep.
- **CANNOT:** analog PHY behavior. Xilinx SERDES/IDELAY in a7ddrphy do not simulate, so read-eye / DQS-gate / pin-LOC problems are silicon-only.

8 Troubleshooting

8.1 Current board state

Sim timing is closed; on-board DDR2 runs but is **gated by PHY calibration**. A clean read-leveling eye is found only at **t_phy_wrlat = 0** (bitslip 0, taps 0–11, centre tap 5). The default of t_phy_wrlat = 4 used by --simple / --char breaks the write path (“no passing tap”).

Known-good board config: t_phy_wrlat = 0, t_rddata_en = 6, rd_phase = 0, rddata_delay = 8, cmd_delay = 1.

After the runtime page-policy fix, OPEN vs CLOSE page policy moved read throughput from ~12.7 MB/s to ~112 MB/s (8.8×) and write to ~44 MB/s (3.5×). For reference, LiteDRAM on the same board reaches ~600 MB/s peak (300 MT/s memtest).

8.2 Sticky errors look like a wedge

STATUS.any_error is **sticky** and is cleared by CTRL.clear_stats, **not** soft_reset. A single failed read latches rd_error; every subsequent engine wait then returns False and looks like a hardware wedge that isn't one. Clear stats between phases. Note pumice_master.wait_engine() waits on **one** engine (write-then-read is phased); a “wait for both” would hang.

8.3 Known bugs / bounds

Known issues and bounds

Symptom	Cause / workaround
Read engine wedges at ~4790 txns	rd_cmd_cam ceiling — keep board runs $\text{txn} \leq 1024$ for clean completion; the perf rate is still valid at scale
row_major wedges immediately	every burst is a row-miss — needs the wedge fix before it can be characterized

Symptom	Cause / workaround
Intermittent 1–11 beat mismatches at scale	clean at $\text{txn} \leq 1024$
col_major stride overflow	stride $16 \text{ KB} \times \text{txn} > 8192$ exceeds 128 MB → address alias; host bounds col_major to $\text{txn} \leq 8192$
synth_mask_obs / lookahead_max_obs read 0	init never set CTRLR_CAP — advertise capability first

8.4 Cannot find the board

The USB-UART re-enumerates across reboots/replugs. `--port auto` (default) probes each `/dev/ttyUSB*`, reads `BUILD_ID`, and picks the one answering `0x44445232`. If none respond it raises “no pumice DDR2 char harness responded ... is the board powered and programmed?” — confirm the bitstream is flashed and the FTDI cable is free.

8.5 a7ddrphy is not simulatable

Verilator cannot model `OSERDESE2` / `ISERDESE2` / `IDELAYE2`, so sim connects at the DFI level with `DFISlavePHY` + `MemoryModel`. Read-eye / DQS-gate / pin-LOC problems are silicon-only. Regenerating the real PHY requires **Python 3.10 only** (migen’s CSR-name tracer breaks on 3.11/3.12):

```
uv venv --python 3.10 /tmp/litex-venv310 && source /tmp/litex-venv310/bin/activate
uv pip install migen "litex==2024.12" "litedram==2024.12" pyyaml
cd projects/fpga-systems/NexysA7/pumice/build-perf
python3 bin/elaborate_a7ddrphy.py --out
rtl-vivado/a7ddrphy/a7ddrphy_generated.v
python3 bin/elaborate_a7ddrphy.py --dump-csr-map      # CSR offsets
for firmware
```

8.6 debug_sram trace ring is disabled on this build

`DEBUG_SRAM_WORDS` was shrunk `32768` → `512` — the full ring needed ~44 K distributed-RAM cells (2.4× over the 100T budget) and blocked `place_design`. The 256-KB address *window* is unchanged (accesses > 2 KB alias into the ring); the ring is effectively unused here. Raise it on a larger device.

8.7 LiteDRAM baseline flow

`flows-litedram-uart` is wired and lint-clean but board bring-up is gated. The default `./regen.sh` builds an empty BIOS ROM (`--no-compile-software`) so the core never asserts

`init_done` — use `./regen.sh --bios` (needs `riscv-gcc`). The host variant (`harness_csr` at base 0, no pumice APB cfg) is still TODO.